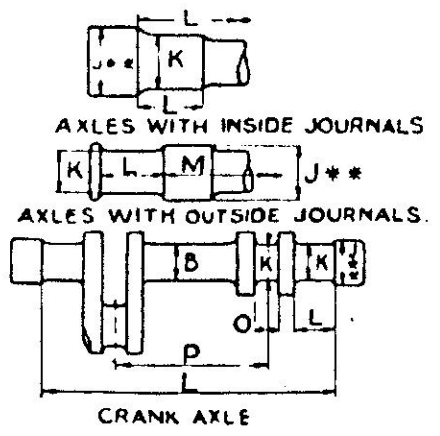
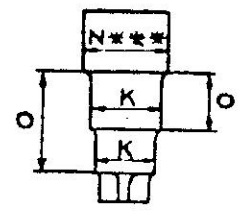
$^{+\cdot002}_{\cdot000}$	**B**	Diameter of seating for eccentric sheaves.
See Note	**J**	* * Diameter at wheel seat.
$^{\cdot000}_{-\cdot002}$	**K**	Diameter of journal.
$^{+\cdot010}_{\cdot000}$	**L**	Distance over inside journals *and* length of journal.
$^{\cdot000}_{-\cdot010}$	**M**	Distance between journals (Axles with outside journals).
$^{+\cdot010}_{-\cdot010}$	**P**	Centres of cranks.



AXLES

$^{\cdot000}_{-\cdot002}$	**K**	Diameter of crank pin where forming bearing for coupling and connecting rods.
See Note	**N**	* * * Diameter of crank pin inside wheel boss.
$^{+\cdot008}_{\cdot000}$	**O**	Crank pin bearing lengths for coupling and connecting rods *and* overall length of these portions.



CRANK PIN

J * * **Cast Steel Wheels.** For all sizes of axles a pressing-in allowance of ·001″ to ·002″ per inch of axle diameter is to be added to the nominal diameter, *e.g.*, an axle 10″ diameter must be made between the limits of 10·010″ and 10·020″. The actual allowance chosen must be such as to give a pressing-in tonnage of 8 to 12 tons per inch of axle diameter. Where axles are designed for inside roller bearings the end of the axle is to be at least $\frac{1}{8}$″ smaller in diameter than the journal in order that the pressing-in allowance may be left on the axle without interfering with the fit of the roller race. It is recommended that wheel centres be bored parallel in all cases, and that axles may be parallel or taper as desired; if tapered, the taper is not to exceed 1 in 500.

Cast Iron Wheels. For all sizes of axles a pressing-in allowance of ·0005″ to ·0015″ per inch of axle diameter is to be added to the nominal diameter, *e.g.*, an axle 10″ diameter must be made between the limits of 10·005″ and 10·015″. The actual allowance chosen must be such as to give a pressing-in tonnage of 6 to 8 tons per inch of axle diameter. Where axles are designed for inside roller bearings, the end of the axle is to be at least $\frac{1}{8}$″ smaller in diameter than the journal, in order that the pressing-in allowance may be left on the axle without interfering with the fit of the roller race. It is recommended that wheel centres be bored parallel in all cases and that axles may be parallel or taper as desired; if tapered the taper is not to exceed 1 in 750.

N * * * **Cast Steel Wheels.** For all sizes of crank pin a pressing-in allowance of ·001″ to ·002″ per inch of crank pin diameter is to be added to the nominal diameter; *e.g.*, a crank pin 4″ diameter must be made between the limits of 4·004″ and 4·008″. The actual allowance chosen must be such as to give a pressing-in pressure of 8 to 12 tons per inch diameter of crank pin when the latter is pressed into the wheel centre. It is of great importance that in all new designs the portion of the crank pin inside the wheel centre shall be at least $\frac{1}{8}$″ larger in diameter than the journal diameter to allow of the pressing-in allowance being left on the crank pin without leaving a sharp shoulder when the journal portion joins wheel seat. In the case of existing designs of crank pins where wheel seat portion and journal portion are the same size the pressing-in allowance shall be deducted from the hole in the wheel seat and not added to the crank pin.

Cast Iron Wheels. For all sizes of crank pin a pressing-in allowance of ·0005″ to ·0015″ per inch of crank pin diameter is to be added to the nominal diameter; *e.g.*, a crank pin 4″ diameter must be made between the limits of 4·002″ and 4·006″. The actual allowance chosen must be such as to give a pressing-in pressure of 6 to 8 tons per inch diameter of crank pin when the latter is pressed into the wheel centre. It is of great importance that in all new designs the portion of the crank pin inside the wheel centre shall be at least $\frac{1}{8}$″ larger in diameter than the journal diameter, to allow of the pressing-in allowance being left on the crank pin without leaving a sharp shoulder where journal portion joins wheel seat. In the case of existing designs of crank pins where wheel seat portion and journal portion are the same size the pressing-in allowance shall be deducted from the hole in the wheel seat and not added to the crank pin.